

Software Testing & Quality Assurance

Lab Report: Test Case Design using EP and BVA

Project Title: Improved Data Leakage Detection System
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Techniques Applied: Equivalence Partitioning (EP) & Boundary Value Analysis (BVA)

1. System Under Test: Improved Data Leakage Detection System

The Improved Data Leakage Detection System is a secure enterprise file-sharing platform designed to provide traceability and accountability. The system allows administrators to upload confidential files, generates unique recipient-specific copies using steganographic tracking tokens, and enables forensic leak analysis to attribute compromised files to their original recipients.

2. Requirement R1 — Secure File Upload and Format Validation [FR-1]

The system shall validate file formats during upload, supporting only `.txt`, `.jpg`, and `.bmp` formats for steganographic tracking. All other file formats must be strictly rejected with a descriptive error message.

Designed Test Case: Format Validation (Equivalence Partitioning)

Field	Details
Test Case ID	TC-DLD-EP-01
Test Scenario	Verify that the system rejects unsupported file formats such as <code>.pdf</code> during file upload using Equivalence Partitioning (Requirement FR-1).
Preconditions	- Administrator account is active and logged into the portal. - Administrator has navigated to the File Upload page (<code>/admin/upload</code>). - System is in an idle state ready to receive upload requests.
Test Steps	- Navigate to the File Upload module. - Click on the “Browse File” button. - Select an unsupported format file: <code>financial_statement.pdf</code>. - Click on the “Upload & Distribute” button.
Input Data	- Target File: <code>financial_statement.pdf</code> - MIME Type: <code>application/pdf</code> - File Size: <code>2.4MB</code>
Expected Result	System blocks the upload and displays an error message: <i>“Invalid file format. Only .txt, .jpg, and .bmp files are supported for steganographic tracking.”</i> File is not stored on server storage.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates invalid partition P4 (Disallowed format) using Equivalence Partitioning (EP) technique.

3. Requirement R2 — File Upload Size Limitation [FR-1 / NFR-3]

To preserve server processing performance and guarantee reliable steganographic embedding, the maximum allowed file upload size is **50 MB** (52,428,800*Bytes*). Any file exceeding 50 MB must be rejected. Empty files (0*Bytes*) are also disallowed.

Designed Test Case: File Size Limit (Boundary Value Analysis)

Field	Details
Test Case ID	TC-DLD-BVA-01
Test Scenario	Verify that the system rejects file uploads that exceed the 50 MB upper size limit using Boundary Value Analysis (Requirement FR-1 / NFR-3).
Preconditions	1. Administrator account is active and logged in with upload permissions. 2. Test file <code>large_dataset.txt</code> of size 50.1<i>MB</i> (52,533,658<i>Bytes</i>) is prepared on the client device.
Test Steps	1. Open the File Upload module (/admin/upload). 2. Select the file <code>large_dataset.txt</code> (50.1<i>MB</i>). 3. Click on the “Upload & Process” button.
Input Data	• Filename: <code>large_dataset.txt</code> • File Size: 50.1<i>MB</i> (52,533,658<i>Bytes</i>)
Expected Result	System halts upload at backend middleware, responds with HTTP 413 Payload Too Large, and displays error: “ <i>Upload failed: File size exceeds the maximum limit of 50 MB.</i> ” File is not stored.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates boundary value 50 <i>MB</i> +0.1 <i>MB</i> (over-limit threshold) using Boundary Value Analysis (BVA) technique.

4. Requirement R3 — Target Agent Concurrent Allocation Count [FR-2]

When distributing a confidential document, the Administrator can assign between **1 and 20 authorized agents** concurrently in a single allocation cycle. Attempting to assign 0 agents or more than 20 agents must be rejected by the system.

Designed Test Case: Agent Batch Allocation Limit (Boundary Value Analysis)

Field	Details
Test Case ID	TC-DLD-BVA-02
Test Scenario	Verify that the system blocks file distribution when selected agent count exceeds the 20-agent batch limit using Boundary Value Analysis (Requirement FR-2).
Preconditions	1. Administrator has uploaded a valid document strategy_doc.txt. 2. System database contains at least 25 registered active agents. 3. Administrator is on the Target Agent Allocation interface.
Test Steps	1. Navigate to the “Target Agent Allocation” screen. 2. Select 21 agent checkboxes from the active agents list. 3. Click on the “Proceed to Steganographic Distribution” button.
Input Data	• Target File: strategy_doc.txt • Selected Agent Count: 21 agents
Expected Result	System blocks the distribution workflow and displays validation alert: <i>“Batch distribution limit exceeded: Maximum 20 recipients permitted per allocation cycle.”</i> No steganographic copies are generated.
Actual Result	(To be completed during test execution)
Status	(Pass / Fail)
Remarks	Validates boundary value $20+1 = 21$ agents (over-capacity threshold) using Boundary Value Analysis (BVA) technique.